

Quiz 3 Section 1

Student Name: _____

Student ID: _____

A manufacturer produces LED bulbs; each bulb has probability $p = 0.03$ of being defective.

Tasks (5 points each)

1. Write down the probability distribution for the number X of successes in n independent trials, where p is the probability of success in a single trial (give the probability mass function).

2. Find the probability that exactly 2 bulbs are defective in a batch of 15 using the distribution given in Task 1.

3. Find the probability that at most 1 bulb is defective in a batch of 15 using the distribution given in Task 1.

4. Using the Poisson approximation, estimate the probability that exactly 2 bulbs are defective when $n = 150$ and $p = 0.03$.